

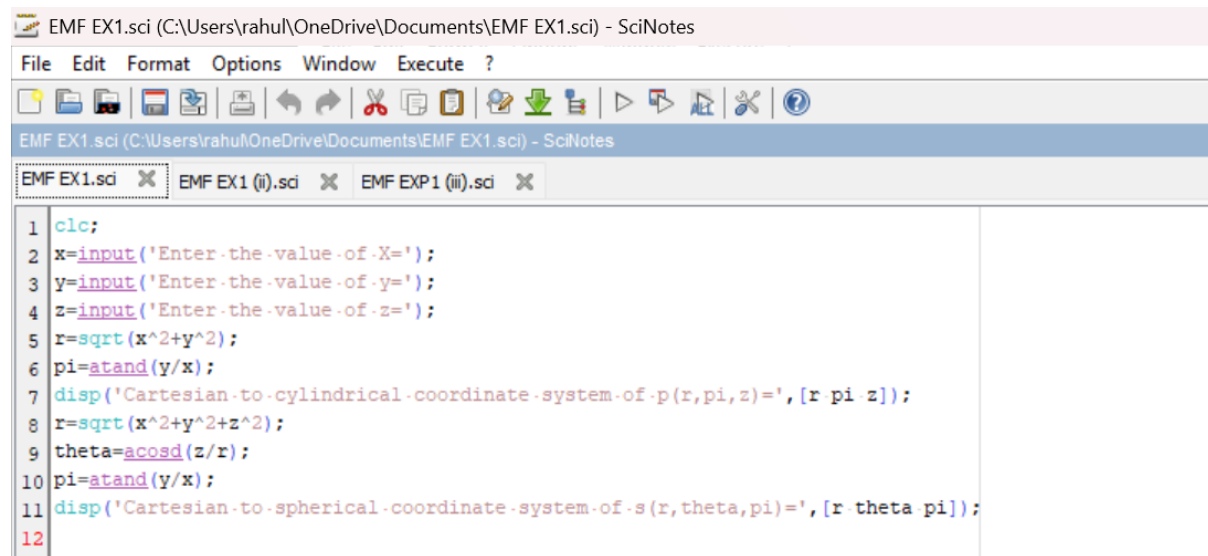
RAHUL R – 192412255

ECA0502 EMF

EXP 1 TESTCASE 1

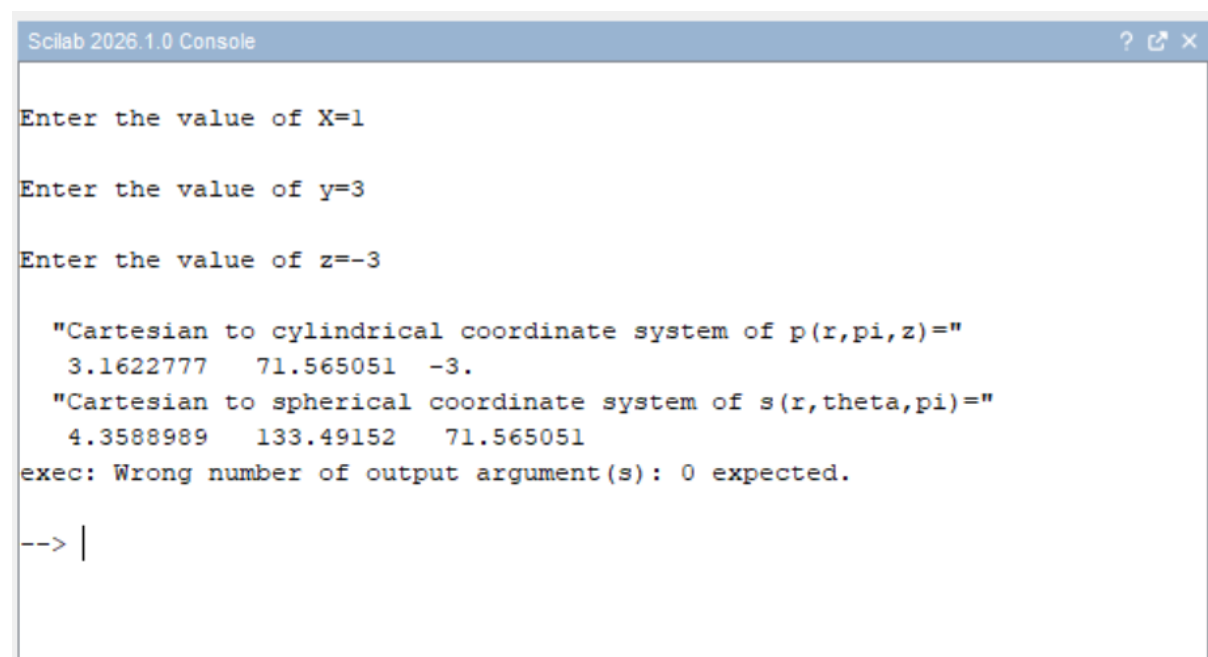
Transformation between Cartesian, Cylindrical and spherical coordinate system

PROGRAM:



```
EMF EX1.sci (C:\Users\rahu\OneDrive\Documents\EMF EX1.sci) - SciNotes
File Edit Format Options Window Execute ?
EMF EX1.sci (C:\Users\rahu\OneDrive\Documents\EMF EX1.sci) - SciNotes
EMF EX1.sci x EMF EX1 (ii).sci x EMF EXP1 (iii).sci x
1 clc;
2 x=input('Enter the value of X=');
3 y=input('Enter the value of y=');
4 z=input('Enter the value of z=');
5 r=sqrt(x^2+y^2);
6 pi=atan2(y/x);
7 disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=[r pi z]');
8 r=sqrt(x^2+y^2+z^2);
9 theta=acosd(z/r);
10 pi=atan2(y/x);
11 disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]');
12
```

OUTPUT:



```
Scilab 2026.1.0 Console
Enter the value of X=1
Enter the value of y=3
Enter the value of z=-3

"Cartesian to cylindrical coordinate system of p(r,pi,z)="
3.1622777 71.565051 -3.
"Cartesian to spherical coordinate system of s(r,theta,pi)="
4.3588989 133.49152 71.565051
exec: Wrong number of output argument(s): 0 expected.
--> |
```

OBSERVATION:

Experiment's observation

EXP: 1

Test case (i) $\Rightarrow x=1, y=3, z=-3$

Cartesian $\rightarrow$ cylindrical

$$\text{radius} \Rightarrow r = \sqrt{x^2 + y^2} = \sqrt{1^2 + 3^2} = \sqrt{1+9} = \sqrt{10}$$

$$r = 3.1625$$

$$\text{Angle} \Rightarrow \phi = \tan^{-1}(y/x) \Rightarrow \tan^{-1}\left(\frac{3}{1}\right)$$

$$\phi = 71.565^\circ$$

$$(r, \phi, z) = (3.162, \dots)$$

$$z = z$$

$$z = -3$$

Cartesian $\rightarrow$ spherical

$$\text{radius} \Rightarrow r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 3^2 + (-3)^2} = \sqrt{19}$$

$$r = 4.3589$$

$$\theta = \cos^{-1}(z/r) = \cos^{-1}(-3/4.3589)$$

$$\theta = 133.492^\circ$$

$$\phi = \tan^{-1}(y/x)$$

$$\phi = 71.565^\circ$$

$$(r, \phi, z) = (3.1625, 71.565, -3)$$

$$(r, \theta, \phi) = (4.3589, 133.492, 71.565^\circ)$$